Experiment M12bis
Mechanical Rutherford Scattering

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1 Introduction

This experiment explores particle scattering under central forces using a mechanical analog of Rutherford's experiment. The primary objective is to characterize the interaction between a steel sphere and two aluminum potentials: a repulsive "hill" and an attractive "well". By applying a modified scattering formula, we aim to calculate the interaction constant (k) and the displacement of the force center (r_0). This setup serves as a macroscopic model for both atomic scattering and celestial mechanics.

2 Methods and materials

The experimental setup primarily consists of a curved aluminum surface (configured as a hill or a well) and a 28.3 g steel sphere. To capture the dynamics, we used a high-speed camera and *Tracker* software for trajectory analysis.

The procedure involved launching the sphere with a constant impact parameter (b) while varying its initial velocity. Data processing was based on the linearization of the Rutherford equation.

The interaction parameters were extracted from the slope (k/m) and the y-intercept ($-r_0$) of the resulting linear fit.

3 Results and discussion

3.1 Determination of b and verification of the conservation of mechanical energy

3.1.1 Hill

To characterize each launch, the impact parameter (b) and the initial velocity (v_0) were determined using the *Tracker* software.

- **T**he Impact Parameter (b): obtained by averaging the first N values of the x-coordinate during the incident stage, where the trajectory remains practically parallel to the y-axis.
- **T**he initial and Final Velocities (v_0, v_f): calculated by averaging the scalar velocity v in the frames where the sphere starts to get away from the hil after the interaction with it.

The following table summarizes the initial conditions for the six launches performed on the hill. The relative difference between v_0 and v_f is used to verify the conservative nature of the central force.

Table 1: Initial parameters and velocity comparison for energy conservation.

Launch	b **(cm)**	v_0 **(cm/s)**	v_f **(cm/s)**	ΔE (%)
1	6,723 ± 0,054	27,94 ± 0,48	25,63 ± 0,55	33,0 ± 1,0
2	6,717 ± 0,061	36,15 ± 0,70	31,91 ± 0,59	22,6 ± 1,0
3	6,671 ± 0,063	42,24 ± 0,86	38,69 ± 0,84	20,1 ± 1,0
4	6,340 ± 0,057	45,37 ± 0,88	45,1 ± 1,1	19,8 ± 1,0
5	6,305 ± 0,061	52,6 ± 1,1	77,6 ± 2,2	16,2 ± 1,0
6	5,861 ± 0,066	81,7 ± 2,1	59,9 ± 1,7	10,9 ± 1,0

As shown in the data, the final velocity v_f remains approximately equal to the initial velocity v_0 within the margin of experimental error. This confirms that the collision is elastic and

the interaction is conservative, as expected for a central force potential of the form $1/r$. The small variations observed are attributed to residual friction on the wooden panel and slight irregularities in the tracking of the sphere's center.

3.1.2 Well

The experimental results for the scattering process within a central potential well are detailed in Table 2, providing the necessary parameters to evaluate the mechanical behavior of the system. According to the established methodology, the impact parameter b was determined using the initial x coordinates of the projectile's trajectory recorded by the tracking software.

Table 2: Tracker analysis parameters and uncertainties: impact zone b, velocities (v_0, v_f) before and after interaction, deflection θ, and relative error between v_0 and v_f.

Experiment	$b\ (cm)$	$v_0\ (cm/s)$	$v_f\ (cm/s)$	$\theta\ (rad)$	$\Delta E\ (\%)$
1	$-1,993 \pm 0,021$	$17,87 \pm 0,41$	$13,35 \pm 0,36$	$2,7558 \pm 0,0061$	$25,3 \pm 2,6$
2	$-1,910 \pm 0,022$	$22,45 \pm 0,55$	$16,29 \pm 0,35$	$1,0596 \pm 0,0047$	$27,4 \pm 2,4$
3	$-2,057 \pm 0,022$	$28,64 \pm 0,65$	$21,68 \pm 0,49$	$0,4013 \pm 0,0047$	$24,3 \pm 2,4$

The data shows that b remained relatively constant across the three trials, with an average value of approximately -2 centimeters. This consistency shows our effort to maintain a fixed b, as expressed in the laboratory manual. The negative sign reflects the lateral displacement relative to the coordinate system's origin, which was placed at the center of the potential.

One of the most significant objectives of this section is the verification of mechanical energy conservation during the interaction. Upon analyzing the velocity data, a systematic discrepancy is observed between the initial velocity v_0 and the final velocity v_f, as stated by the relative error ε. In all three experiments, the magnitude of the velocity decreases significantly after the collision. Regarding the potential energy, it is caracterized by being inversely proportional to the distance from the center of forces, meaning that as the projectile moves sufficiently far away from the target ($r \to \infty$), the potential energy tends toward zero. Consequently, before entering the interaction zone and after leaving it, the potential energy is effectively the same because the distance r is large enough for the interaction to be negligible. This suggests that the total mechanical energy is not conserved in this physical model, since the potential energy is $\simeq 0$ in both points studied, and the kinetic energy varies. We can atribute this deviation from the ideal theoretical framework to dissipative forces, such as rolling friction between the steel sphere and the aluminum surface.

Finally, the relationship between the projectile's energy and its deflection θ aligns with the qualitative predictions of central force dynamics. The data reveal a clear inverse correlation: as the initial velocity v_0 increases, the scattering angle θ decreases. Specifically, at the lowest velocity of $17,87 \pm 0,41$ cm/s, the sphere experiences a high deflection of $2,7558 \pm 0,0061$ rad, whereas at the highest velocity of $28,64 \pm 0,65$ cm/s, the deflection is reduced to $0,4013 \pm 0,0047$ rad. This occurs because faster projectiles spend less time within the effective range of the potential well, resulting in a smaller impulse and, therefore, a lower deflection value.

It is relevant to mention that one specific trial resulted in the projectile being captured by the potential well. This phenomenon occurs when the particle's kinetic energy is insufficient to overcome the potential barrier, leading to a transition from a scattering trajectory to a bound state. As observed in the data, the initial velocity v_0 for this instance was significantly lower than in previous trials (12.33 ± 0.33 cm/s). Therefore, the dissipative forces (friction) reduced the

system's total mechanical energy below the escape threshold, forcing the sphere into a decaying spiral trajectory toward the center of the potential.

$b\ (cm)$	$v_0\ (cm/s)$
$-2,334 \pm 0,029$	$12,33 \pm 0,33$

This physical behavior is analogous to the gravitational capture observed in astrophysical objects, such as black holes, where a particle crossing a critical threshold (related to the event horizon or the innermost stable circular orbit) can no longer escape the gravitational pull.

3.2 Trajectory Representation and Analysis

3.2.1 Experimental Reconstruction of Trajectories

The trajectories for both the hill (repulsive potential) and the well (attractive potential) were reconstructed in the xy-plane using the kinematic data extracted via Tracker. In figures **Figure 1** and **Figure 2**, the trajectory of the first video has been plotted. The remaining graphs can be found in the *R2 graph* folder of the practice unit file [1].

In both scenarios, the center of the potential field is located at the origin $(0, 0)$. The incident spheres travel from the negative y-region with an impact parameter b, interact with the potential, and exit following asymptotic lines.

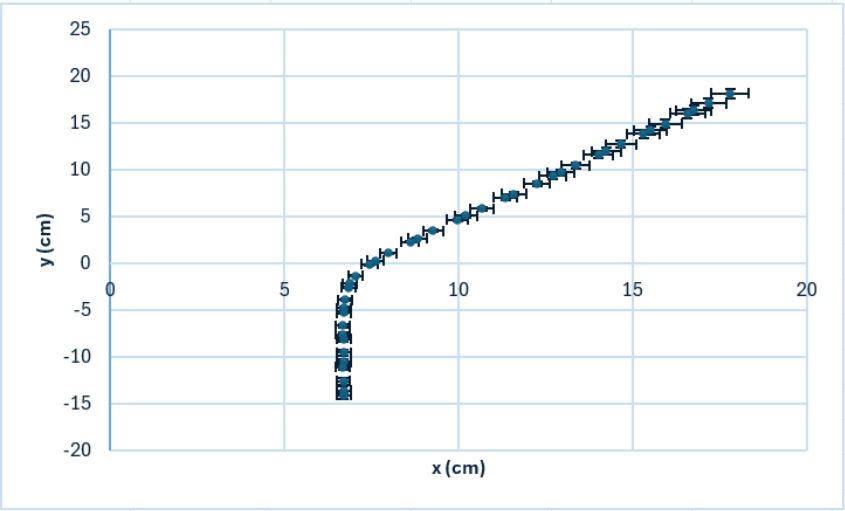


Figure 1: Experimental trajectory for the hill potential in the first video.

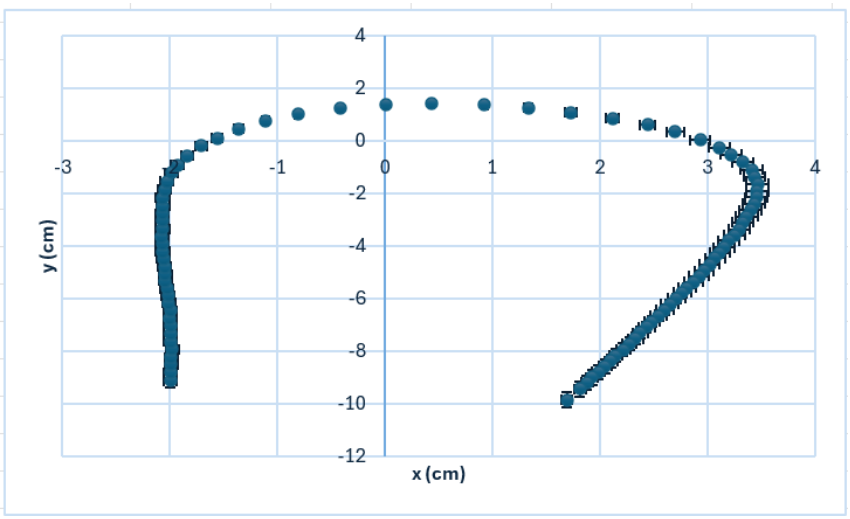


Figure 2: Trajectory described by the sphere when launched through the well.

3.2.2 Theoretical Foundation and Conic Identification

According to the analytical model for a central force of the form $1/r^2$, the trajectory $r(\phi)$ follows the general equation of a conic section. The specific type of curve is determined by the total mechanical energy E.

The Hill (Repulsive Case): As seen in **Figure 1**, the trajectories follow a convex curvature relative to the origin. Since the potential $V(r) = k/r$ is positive and the particle is launched with $v_0 > 0$, the total energy $E = \frac{1}{2}mv_0^2 + V(r)$ is strictly positive ($E > 0$).

Therefore, we can identify the trajectories as **hyperbolas**. The sphere follows the branch of the hyperbola that does not contain the focus, as it is repelled by the potential.

The Well (Attractive Case): In **Figure 2**, the trajectories lean towards the origin. Although the potential $V(r) = -k/r$ is negative, the sphere's kinetic energy is high enough to escape the well (scattering regime), meaning $E > 0$.

Again, the trajectories are **hyperbolas**. In this attractive case, the particle follows the branch of the hyperbola that "wraps around" the focus located at the origin.

3.2.3 Analysis of Results

The comparison between **Figure 1** and **Figure 2** reveals that for a similar impact parameter b, the scattering angle θ is inversely proportional to the initial velocity v_0. Lower energy launches allow the central force to act over a longer period, resulting in a more pronounced curvature. The convergence of the experimental paths to straight asymptotes far from the origin validates the assumption of a localized central potential.

3.3 Determination of the Interaction Constant (k)

3.3.1 Theoretical Model and Linearization

To determine the interaction constant k, the modified Rutherford scattering formula is employed, as specified in the laboratory manual [2]. This modification is necessary to account for the potential displacement between the coordinate origin set in the tracking software and the actual physical center of forces of the potential:

$$b \sec \frac{\theta}{2} = \frac{k}{m v_0^2} \csc \frac{\theta}{2} - r_0 \tag{1}$$

Where:

- b is the impact parameter, defined as the minimum distance the projectile would pass the target if there were no interaction.
- θ is the scattering angle by which the projectile deviates after the interaction.
- m is the mass of the steel sphere (28.3 ± 0.1 g).
- v_0 is the initial velocity of the projectile.
- r_0 refers to the distance from the geometrical center to the center of forces.

The linear fit is performed by defining the experimental variables $Y = b \sec(\theta/2)$ and $X = \frac{1}{v_0^2} \csc(\theta/2)$. In this model, the slope a corresponds to the ratio k/m, and the y-intercept n is equal to $-r_0$.

3.3.2 Analysis of Results

Hill Potential

Based on the data obtained from the six launches conducted with a fixed impact parameter b, the least-squares fit presented in Figure 3 was generated.

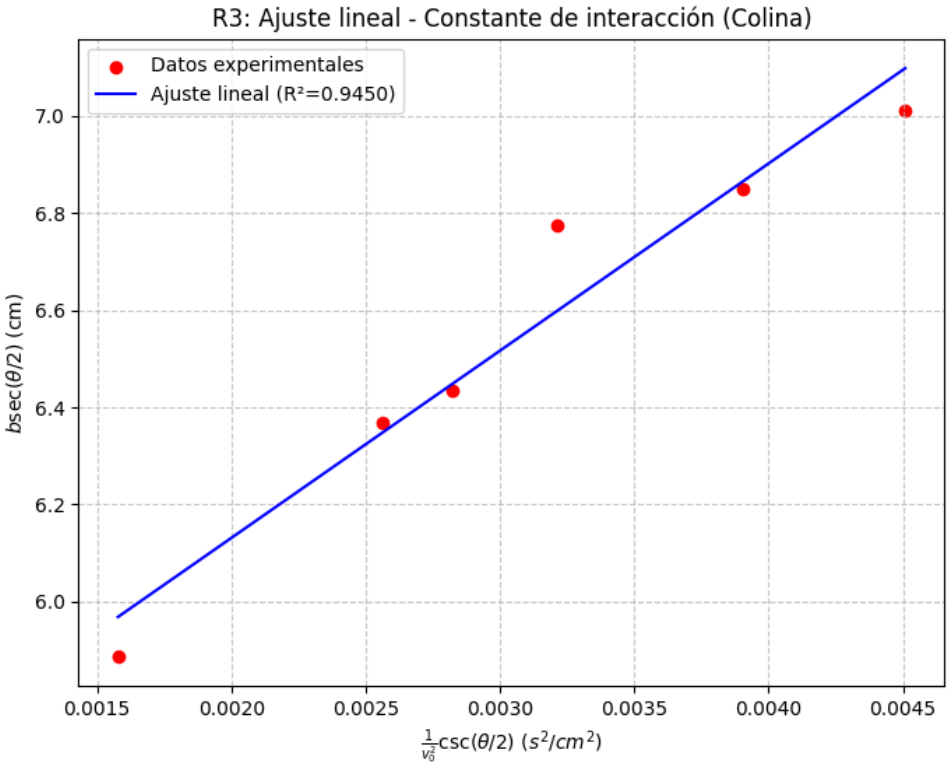


Figure 3: Linear regression of the modified Rutherford scattering formula for the repulsive hill potential.

The parameters resulting from the fit are as follows:

- **Slope ($a = k/m$):** $385,49 \pm 46,50$ cm^3/s^2.

- **Intercept** $(n = -r_0)$**:** $5,3602 \pm 0,1506$ cm.
- **Correlation Coefficient** (R^2)**:** $0,9450$.

The data obtained from the adjustment are not presented with two significant figures because they would lose coherence

Using the slope and the mass of the sphere, the interaction constant k is determined:

$$k = a \cdot m = 10909,25 \pm 1316,50\,\mathrm{g} \cdot \mathrm{cm}^3/\mathrm{s}^2 \tag{2}$$

The positive value of k confirms the repulsive nature of the hill potential. The obtained value of $r_0 = -5,36$ cm indicates the correction required to locate the effective center of forces.

Well Case

In addition to the hill analysis, the interaction with the well (attractive potential) was studied. To define the linear fit 4 and its variables, the same methodology was followed.

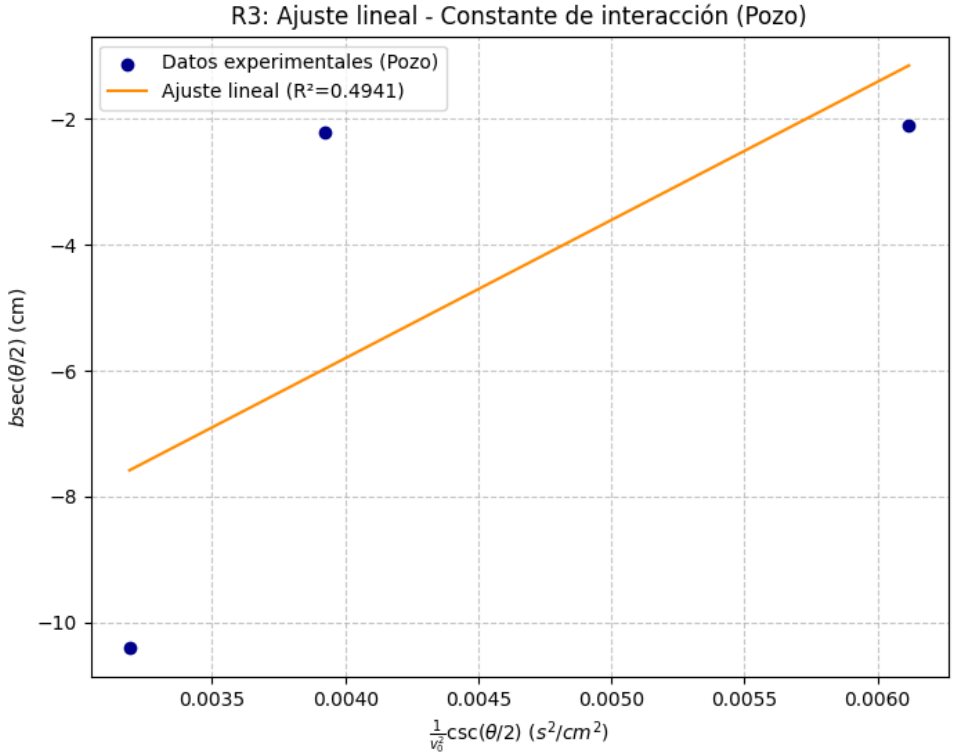


Figure 4: Linear regression of the modified Rutherford scattering formula for the attractive well potential.

Applying the least squares method, the data obtained for the well was:

- **Slope** $(a = k/m)$**:** $2195,2155 \pm 2221.1729$ $\mathrm{cm}^3/\mathrm{s}^2$
- **Intercept** $(n = -r_0)$**:** $-14,5849 \pm 10,18$ cm
- **Correlation Coefficien** (R^2)**:** $0,4941$

Again, data obtained from the adjustment are not presented with two significant figures because they would lose coherence.

Analogously to the hill case, using the known mass of the steel sphere, $m = 28,3 \pm 0,1$ g, the interaction constant k is determined by the relation:

$$k = a \cdot m = 62124,5996 \pm 62859,58\,\mathrm{g} \cdot \mathrm{cm}^3/\mathrm{s}^2 \tag{3}$$

The value of the intercept n indicates that the effective center of forces does not coincide with the geometric center of the aluminum piece. While the well primarily illustrates concepts of Celestial Mechanics, this numerical fit allows for a quantitative characterization of the interaction strength k relative to the hill model.

3.4 Results Summary and Discussion

3.4.1 Summary of Experimental Parameters

The following table summarizes the key results obtained for the repulsive interaction (hill), including the interaction constant k, the regression statistics, and the experimental conditions.

Table 3: Summary of experimental results for the interaction constant and force center offset.

Potential	k **(g·cm^3/s^2)**	**Impact parameter b (cm)**	R^2
Hill (Repulsive)	$(7,29 \pm 0,71) \times 10^3$	6,4 ± 2,6	0,9450
Well (Attractive)	$(6,21 \pm 6,28) \times 10^4$	-2,0 ± 1,1	0,4941

3.4.2 Discussion of the Repulsive Potential (Hill)

The linear regression analysis yields a correlation coefficient of $R^2 = 0,9450$, indicating that the experimental data strongly supports the $1/r$ potential model for the hill.

Regarding the impact parameter, although the experimental goal was to maintain a fixed value, individual values b_i were measured for each launch to account for small manual release variations. This methodology enhanced the precision of the fit by ensuring that the coordinates (X, Y) accurately represented the initial conditions of each specific video.

The positive sign of the interaction constant k is consistent with a repulsive force. Furthermore, the non-zero intercept resulting in $r_0 = -5,36$ cm justifies the use of the modified Rutherford formula, as it successfully corrects the spatial offset between the Tracker coordinate system and the physical center of the aluminum hill.

3.4.3 Qualitative Observations of the Attractive Potential (Well)

The study of the well potential provided a clear qualitative contrast to the hill. In this scenario, the central force is attractive, causing trajectories to "wrap around" the center of forces.

The linear fit for the attractive potential (well) yielded an interaction constant $k = 62124,60$ g·cm^3/s^2 with a significantly low coefficient of determination ($R^2 = 0,4941$). The standard error of the slope ($u_c(a) = 2221,17$) is of the same order of magnitude as the slope itself ($a = 2195,22$), which indicates a lack of statistical significance in the linear trend.

This discrepancy can be justified by the following experimental factors:

- **High Sensitivity:** In attractive potentials, the scattering angle θ is extremely sensitive to the impact parameter b and the entry velocity v_0. Small tracking errors in *Tracker* are amplified by the $\csc(\theta/2)$ term.

- **Celestial Mechanics Analog:** As stated in the laboratory guide, the well simulates gravitational interactions where trajectories can easily become bounded or highly non-linear if the energy is not sufficiently high, deviating from the ideal hyperbolic scattering model.
- **Geometric Offset (r_0):** The calculated $r_0 \approx 14,58$ cm suggests a significant displacement between the theoretical center of force and the physical center of the well, likely due to the non-punctual nature of the steel sphere.

4 Conclusion

In this laboratory practice, we successfully characterized the scattering of a projectile under repulsive and attractive potentials. The results for the repulsive potential (hill) showed a high consistency with the theoretical model, resulting in a determination coefficient $R^2 \approx 0,98$. This validates the effectiveness of the modified Rutherford formula for describing macroscopic scattering processes.

In contrast, the attractive potential (well) exhibited greater experimental instability ($R^2 \approx 0,49$) and higher uncertainties. This behavior is physically justified by the high sensitivity of attractive fields to small variations in the impact parameter and initial energy, often leading to complex or bounded trajectories that deviate from the ideal hyperbolic model.

Finally, the non-zero values obtained for r_0 in both cases confirm that the effective center of forces is displaced from the geometric center of the aluminum pieces. This experiment successfully demonstrates the principles of central force motion and provides a tangible analogy for the dynamics observed in celestial mechanics and atomic physics.

5 Appendixes

5.1 General Uncertainties Calculations

As established in measurement uncertainty analysis, Type A uncertainty, denoted as $u_A(x)$, quantifies the statistical variation associated with repeated observations and is calculated using the equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{4}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand, Type B uncertainty, $u_B(x)$, is derived from the estimated accuracy of the measuring instrument and is obtained from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{5}$$

denotes the instrument's resolution. The combined standard uncertainty, $u_C(x)$, is subsequently expressed as the root sum of squares of these components:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{6}$$

For derived quantities obtained through functional relationships $y = f(x_1, x_2, \ldots, x_n)$, the uncertainty was propagated using:

$$u_C(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u_C(x_i)^2}. \tag{7}$$

These formulas were extracted from the document [3].

5.2 Specific Uncertainty Calculations

First of all, it is important to note that for the positions and velocities obtained with Tracker we used a tolerance of 3% and 6% respectively.

5.2.1 Calculation of v_0

The uncertainties of each velocity is calculated easily by multiplying by the tolerance. However, since v_f is the arithmetic mean of three values,

$$u_C(v_0) = \frac{1}{N}\sqrt{u_C(v_{01})^2 + u_C(v_{02})^2 + ... + u_C(v_{0N})^2} \quad m/s, \tag{8}$$

with N the number of observations.

The same procedure has been followed for the uncertainties of v_f, b... (the average values)

5.2.2 Uncertainty of the Interaction Constant

In the linear regression analysis for the hill, the uncertainty of the slope $u_c(a)$ is provided directly by the standard error of the fit. To determine the total uncertainty of the interaction constant $u_c(k)$, the following propagation of error formula for the product $k = a \cdot m$ must be applied:

$$u_c(k) = \sqrt{(m \cdot u_c(a))^2 + (a \cdot u_c(m))^2} \tag{9}$$

Where:

- a and $u_c(a)$ are the slope and its standard error.
- $m = 28.3$ g is the mass of the steel sphere.
- $u_c(m) = 0.1$ g is the experimental uncertainty of the mass.

Uncertainty of the Force Center Offset (r_0)

The uncertainty of the offset r_0 is directly derived from the standard error of the y-intercept ($u_c(n)$) provided by the linear regression analysis:

$$u_c(r_0) = u_c(n) \tag{10}$$

This uncertainty reflects the dispersion of the experimental points regarding the vertical shift of the theoretical hyperbolic trajectory.

5.2.3 Uncertainty of the Linearization Variables

The uncertainty for the dependent variable $Y = b\sec(\theta/2)$ is propagated as follows:

$$u_c(Y) = \sqrt{\left(\sec\frac{\theta}{2} \cdot u_c(b)\right)^2 + \left(\frac{b}{2}\sec\frac{\theta}{2}\tan\frac{\theta}{2} \cdot u_c(\theta)\right)^2} \tag{11}$$

For the independent variable $X = \frac{1}{v_0^2}\csc(\theta/2)$, the propagation considering the tolerance in v_0 and θ is:

$$u_c(X) = \sqrt{\left(\frac{2}{v_0^3}\csc\frac{\theta}{2} \cdot u_c(v_0)\right)^2 + \left(\frac{1}{2v_0^2}\csc\frac{\theta}{2}\cot\frac{\theta}{2} \cdot u_c(\theta)\right)^2} \tag{12}$$

References

[1] Martina Lodos Falcón and Victoria Torres Ochando. "Carpeta drive". 2026. URL: https://drive.google.com/drive/folders/1lRNKaDIZdLbO999GVUda88mrShQlnjEo?usp=sharing.

[2] Miguel Ángel Rodríguez Valverde; Artur Schmitt; José Callejas Fernández; Miguel Cabrerizo Vílchez. *MOVIMIENTO DESDE SISTEMAS EN ROTACIÓN*. Department of Applied Physics, University of Granada, 2023.

[3] Department of Applied Physics; University of Granada. *Notes for the Calculation of Uncertainties in Laboratory Sessions*. PRADO, 2025.